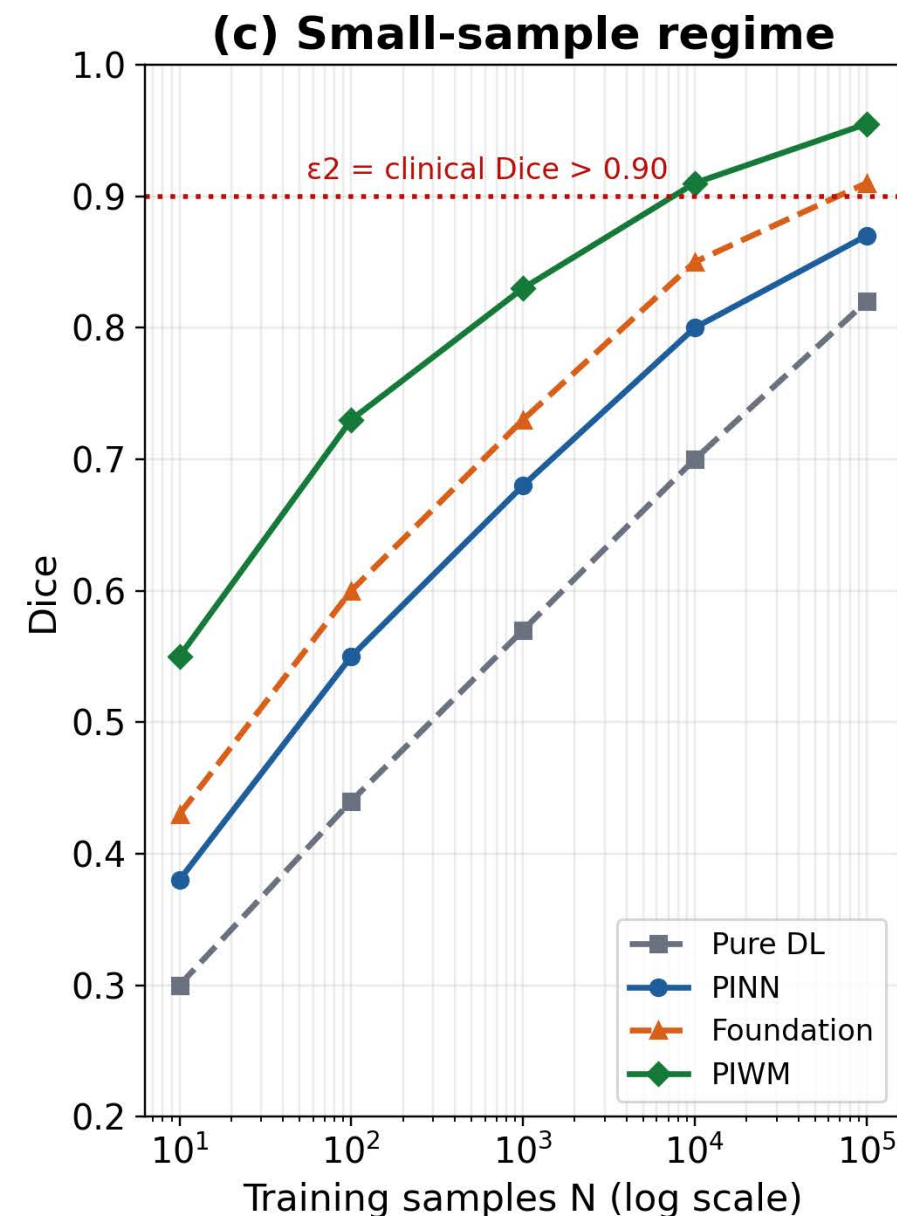
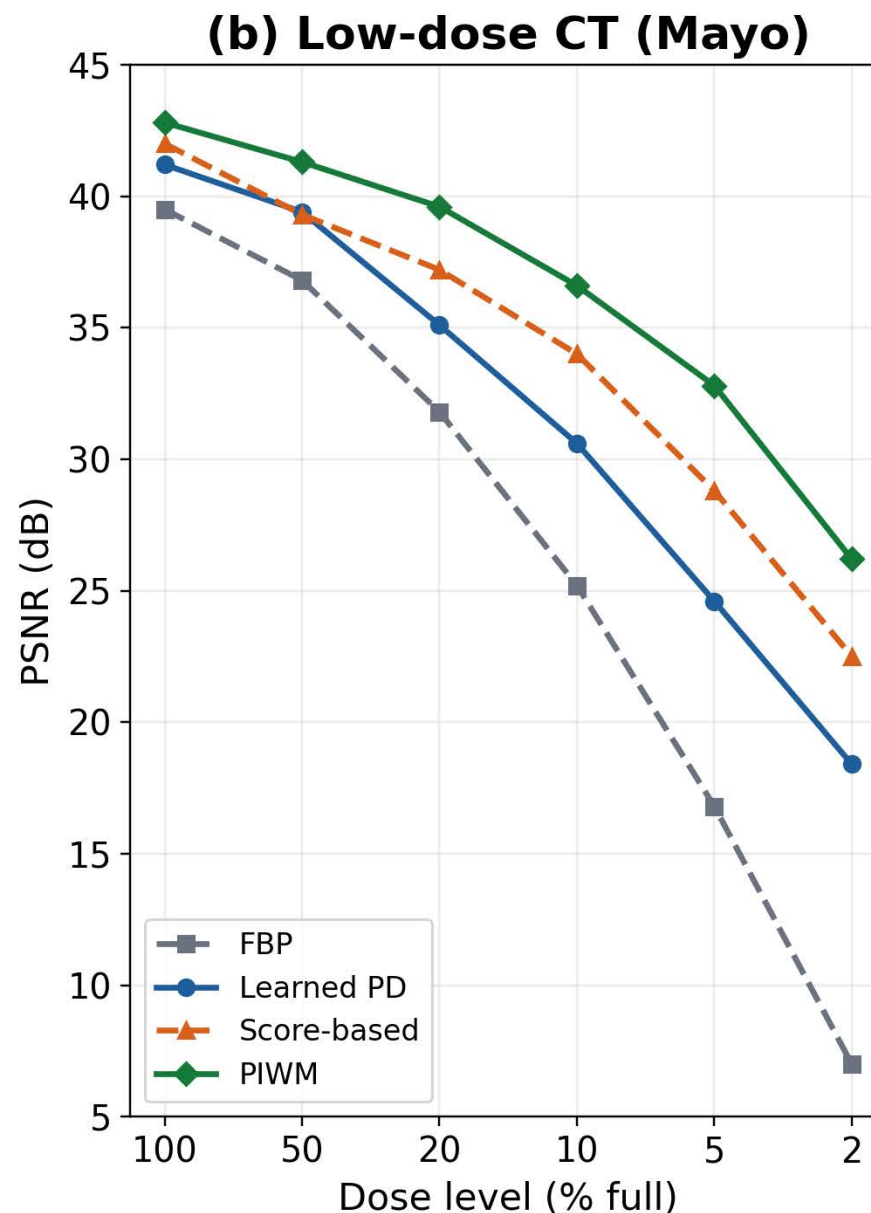
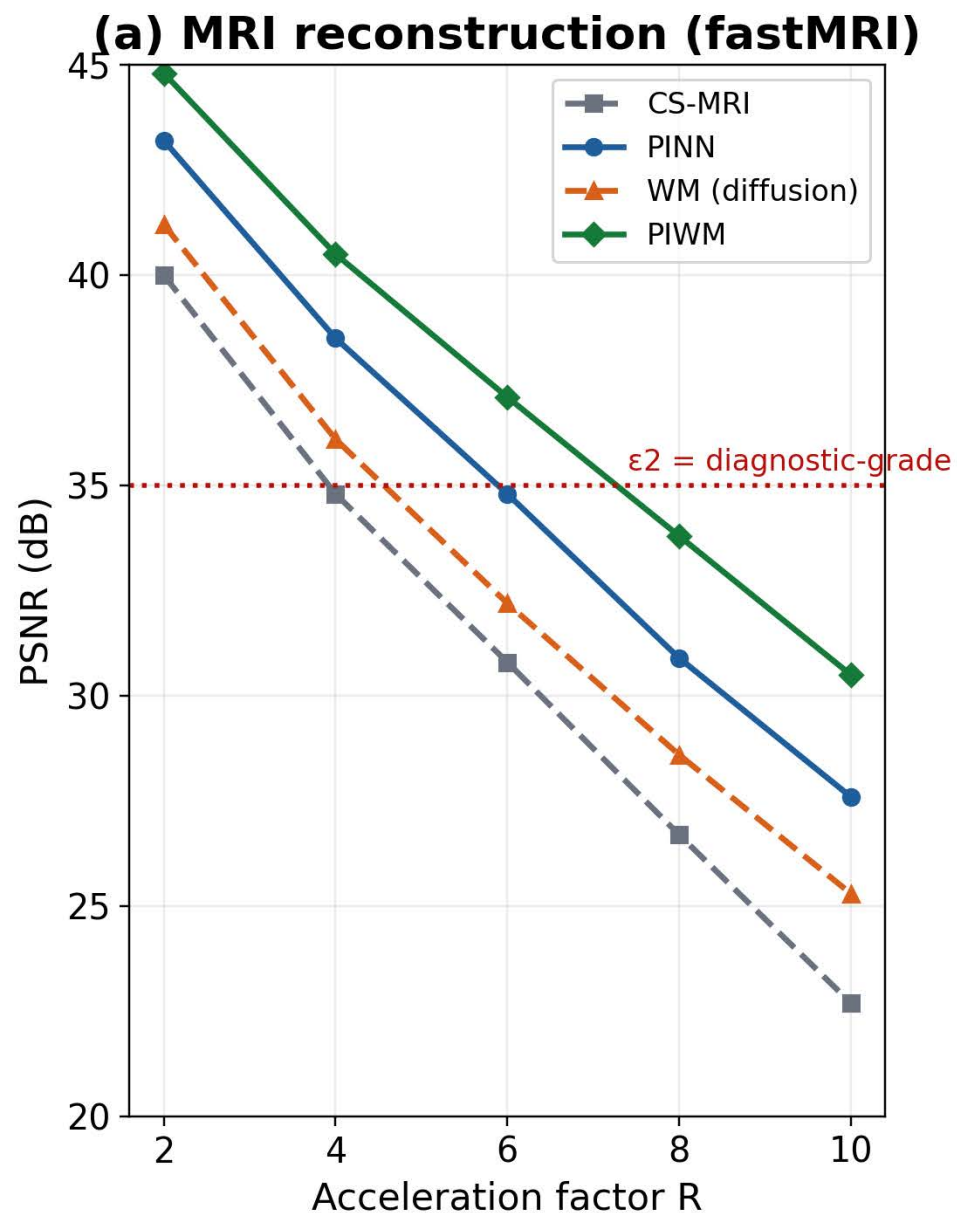


Quantitative literature benchmarks



PIWM consistently outperforms pure-PINN and pure-world-model baselines; thresholds are reached earlier across settings.